Multi-Component Style Vectors with Norm-Preserving Steering for Personalized Text Generation: Reproduction and Extension of StyleVector

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***Abstract*—Large language models write in one average style, yet users want text in their own voice. StyleVector (Zhang et al., ACL 2025) represents a user's style as a single mean-difference direction in an LLM's activation space and adds it to the hidden states during generation, without retrieval or training. No official code was available, so we reimplemented it from the paper and evaluated it on three LaMP tasks (news headlines, scholarly titles, tweet paraphrasing) with LLaMA-2-7B-chat. At $n = 1{,}000$ users per task it improves all six task–metric pairs, in the paper's direction and with similar magnitude (for example $+23.1\%$ ROUGE-L on LaMP-5 against the paper's $+25.8\%$). We then proposed and evaluated two extensions that target limitations the paper names: multi-component style vectors (top-k PCA and a shared-basis variant) and norm-preserving steering. Across five experimental runs and a full 2×2 ablation, with sweeps over k, layer and strength and with global-vector, shrinkage and random-direction controls, neither extension improved on the paper's method. The shared-basis vector was statistically indistinguishable from Mean Difference in all six paired comparisons, and norm-preserving variants scored lower in three of six. The study also shows that evaluation choices matter: output cleaning alone raised baseline ROUGE-L by 14–51%. We also built a live demonstration system, and we document every deviation from the paper and every use of AI assistance.**

***Index Terms*—Personalized text generation, activation steering, style vectors, reproducibility, ablation study, LaMP.**

I. Introduction

LARGE language models, built on the Transformer architecture [1] and trained with deep learning at scale [2], produce a single, average writing style. Real deployments increasingly need text that sounds like a specific person: emails, reviews, headlines and posts in the user's own voice. Two approaches dominate. Retrieval-augmented generation (RAG) injects the user's past writing into the prompt [3], [4], so latency grows with history size. Parameter-efficient fine-tuning (PEFT) trains a small adapter per user [5], [6], which needs per-user training and storage. Both also entangle *what* a person writes about with *how* they write it.

StyleVector [7] proposes a third route. It contrasts the user's authentic writing with a style-agnostic rewrite of the same content, averages the difference of the model's hidden states into one *style vector*, and adds this vector back during generation. It needs no retrieval and no training, and storage is one vector per user. The authors themselves note two limitations: a single vector may conflate several stylistic dimensions, and steering is unstable when it is too strong.

A. Project criterion and team

We follow the criterion *"Reproduce paper results if code not available"*, combined with a proposed and evaluated improvement. This is a single-member project. **Member 1:** Slok Tulsyan, Roll No. 12342080, slokt@iitbhilai.ac.in. All parts were carried out by the author: literature review, reimplementation, the demonstration system, the extension, all experiments and ablations, and this report.

B. Contributions

1) A from-scratch reimplementation of StyleVector that reproduces the paper's direction and approximate magnitude of improvement on all six task–metric pairs at $n = 1{,}000$ users per task, with every deviation disclosed (Section V-A).
2) An interactive web application that runs the full pipeline live (Section IV).
3) Two extensions, multi-component style vectors and norm-preserving steering, evaluated with a 2×2 ablation over five runs, three controls and sweeps over k, layer and strength (Sections III–VI).
4) A diagnosis of why early reproduction attempts diverged from the paper, including a quantified output-format confound (Sections VI and VII).

II. Related Work

Personalized generation. The LaMP benchmark [8] and its long-form counterpart LongLaMP [9] evaluate personalization from user histories. RAG-based methods retrieve history items by semantic similarity [3]; the StyleVector authors argue that this tends to retrieve content rather than style [7]. PEFT-based methods train per-user adapters [5], [6], [10]. Salemi and Zamani compare the two families for privacy-preserving personalization [4], and the StyleVector authors observe that RAG is more effective with little user history and PEFT with more [7].

Activation engineering. Linear directions in hidden states can steer model behaviour: activation addition [11], contrastive activation addition for Llama 2 [12], representation engineering [13] and in-context vectors [14]. StyleVector applies this idea to a user's writing style [7].

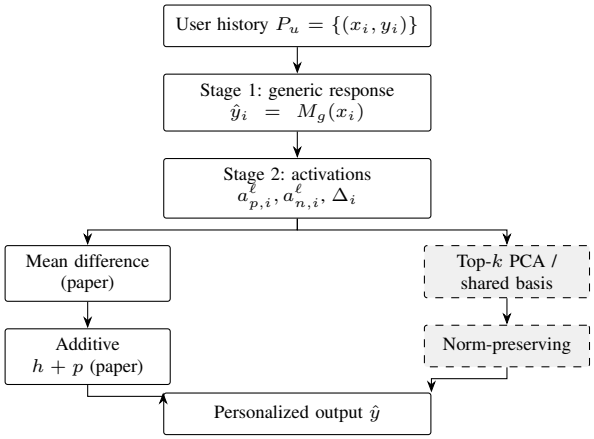


Fig. 1. Pipeline. Solid: original StyleVector stages. Dashed: our extensions (multi-component extraction and norm-preserving steering).

Multiple directions. Multi-directional orthogonal steering has been used to improve truthfulness [15], and sparse autoencoders decompose activations into interpretable features [16]; StyleVector's own limitations section suggests sparse combinations of concept-specific vectors as future work. Our multi-component vectors are a lighter-weight step in that direction, using PCA instead of learned features.

Gap. The original paper evaluates one direction with one fixed additive rule and no ablation of its components. We isolate the contribution of multi-component representation and of norm preservation with controlled comparisons and controls that test whether any gain is personal.

III. Proposed Methodology

Fig. 1 shows the pipeline. Solid boxes are the original StyleVector stages; dashed boxes are our modifications.

A. StyleVector baseline

For user u with history $P_u = \{(x_i, y_i)\}$, a general model M_g produces a style-agnostic response

$$\hat{y}_i = M_g(x_i). \tag{1}$$

With $h_\ell(r)$ the last-token hidden state at layer ℓ for text r, the positive (authentic) and negative (generic) activations are

$$a^{\ell}_{p,i} = h_\ell(x_i \oplus y_i), \qquad a^{\ell}_{n,i} = h_\ell(x_i \oplus \hat{y}_i), \tag{2}$$

and the Mean Difference style vector is

$$s^{\ell}_u = \frac{1}{|P_u|} \sum_i \big(a^{\ell}_{p,i} - a^{\ell}_{n,i}\big). \tag{3}$$

Generation adds the scaled vector at every generated token at one layer:

$$h'_\ell(x)_t = h_\ell(x)_t + \alpha\, s^{\ell}_u. \tag{4}$$

B. Multi-component style vectors

Let $\Delta_i = a^{\ell}_{p,i} - a^{\ell}_{n,i}$. A single mean can mix lexical, syntactic and discourse-level style. We therefore built two multi-component representations.

Per-user top-k PCA. The top-k principal components $\{v_1, \ldots, v_k\}$ of the symmetric set $\{\Delta_i\} \cup \{-\Delta_i\}$ are the leading eigenvectors of $\sum_i \Delta_i \Delta_i^\top$. They are combined as

$$\tilde{s}_u = \frac{\|s_u\|}{\|z\|}\, z, \quad z = \sum_{j=1}^{k} w_j \operatorname{sgn}\big(\langle v_j, s_u\rangle\big)\, v_j, \tag{5}$$

with $w_j \propto \sqrt{\lambda_j}$ (explained variance), $\sum_j w_j = 1$, and rescaling to the mean-difference norm so that a given α is comparable across methods.

Shared-basis projection. A per-user estimate from 15–60 history items in $d = 4096$ dimensions is noisy. We therefore learn a basis from *all other* users' mean vectors m_v (history only, no gold outputs). Let V_r hold the top-r right singular vectors of the matrix of $\{m_v\}_{v \neq u}$ (leave-one-out) and $P_r = V_r^\top V_r$. The user vector is

$$s^{\mathrm{sh}}_u = \frac{\|m_u\|}{\|P_r m_u\|}\, P_r m_u, \qquad r = 16. \tag{6}$$

C. Norm-preserving steering

The paper reports quality collapse at large α. After adding the perturbation $p = \alpha s_u$ (or its multi-component analogue) we rescale to the original activation norm:

$$h'_\ell(x)_t = \|h_\ell(x)_t\| \cdot \frac{h_\ell(x)_t + p}{\|h_\ell(x)_t + p\|}. \tag{7}$$

To first order this removes only the component of p parallel to h. We also use the relative strength $\rho = \|p\|/\|h\|$ in two runs.

D. Controls

Global vector: one non-personalized vector (mean over users) for all users. *Shrinkage:* the user vector shrunk towards the global vector ($\kappa = 10$). *Random direction:* a random vector of the same norm, with norm preservation, at the same layer and strength.

IV. Experimental Setup

A. Data

We use the public LaMP benchmark [8] (https://lamp-benchmark.github.io/): News Headline Generation (LaMP-4), Scholarly Title Generation (LaMP-5) and Tweet Paraphrasing (LaMP-7). LaMP does not release test labels, so we evaluate on the public *dev* split. Each task gives, per user, historical (input, output) pairs and a held-out query with a reference. Histories are capped for compute (15 items in the reproduction; see Table I for the extension runs). Preprocessing is minimal: tokenization by the model's tokenizer and truncation of very long inputs.

Algorithm 1 Steered generation with the extensions

Input: history P_u, query x, layer ℓ, strength α (or ρ), mode $\in$ {mean, PCA-k, shared-r}, NP $\in \{0, 1\}$
for $(x_i, y_i) \in P_u$ **do**
  $\hat{y}_i \leftarrow M_g(x_i)$; $\Delta_i \leftarrow h_\ell(x_i \oplus y_i) - h_\ell(x_i \oplus \hat{y}_i)$
end for
build s_u from $\{\Delta_i\}$ according to mode (Eqs. 3, 5, 6)
for each generated position t (and prompt positions, if enabled) **do**
  $h \leftarrow h_\ell(x)_t$; $h' \leftarrow h + \alpha s_u$
  if NP **then**
    $h' \leftarrow \|h\| \, h'/\|h'\|$
  end if
end for
return greedy decoding with the modified hidden states

B. Model, hardware and software

We use `meta-llama/Llama-2-7b-chat-hf` [17], matching the paper. The reproduction ran on one NVIDIA RTX 3060 (12 GB), much smaller than the paper's 8×RTX 3090 cluster. With 8-bit loading the model used 11.8 of 12.3 GB and hung intermittently, so we switched to 4-bit NF4 quantization [18], a disclosed, hardware-driven deviation. The extension runs used the lab GPU machine (device not recorded in the archives). Software: PyTorch [19], HuggingFace Transformers [20], `bitsandbytes`, `rouge-score`, NLTK, pandas and matplotlib. Activations are extracted and injected with forward hooks, without changing weights.

C. Prompts and decoding

The reproduction uses the paper's Appendix E prompt templates and greedy decoding. The refined pipeline (Run 5) uses chat-templated prompts with LaMP-style wording and a generation limit of 40 tokens.

D. Metrics and statistics

We report ROUGE-L [21] and METEOR [22], as in the paper. For the extension we compute paired bootstrap confidence intervals [23] over users (10,000 resamples, 95% percentile interval) on the mean difference, expressed as a percentage of the no-steering score. No multiple-comparison correction is applied, so isolated significant cells are interpreted with care.

E. Hyper-parameter selection and the five extension runs

In the reproduction, layer ℓ and strength α were chosen by grid search on a held-out validation subset per task, following the paper (final: layer 16, $\alpha = 0.25/0.5/0.5$ for LaMP-4/5/7). The extension was run five times; each run changed the protocol (Table I).

F. Demonstration system

A Flask web application runs the full pipeline interactively. The user selects a task and a person from LaMP and supplies free text or the person's actual held-out query. The application shows the baseline and the steered output, ROUGE-L and METEOR when a reference exists, a reference-free *style alignment* score (cosine similarity between the output's activation and the user's style vector), and the history items that contribute most to the style vector. It reuses the same activation-extraction, Mean Difference and hook code as the batch evaluation, and caches vectors per user.

V. Results and Analysis

A. Reproduction of the paper's method

Table II compares the reproduction at $n = 1{,}000$ users per task with the paper's Table 2. StyleVector improves over the non-personalized baseline on all six task–metric pairs, in the same direction as the paper. Relative gains are within a few percentage points of the paper's in most cells: LaMP-4 $+3.3\%/+3.0\%$ (paper $+3.2\%/+2.5\%$), LaMP-5 $+23.1\%/+10.5\%$ ($+25.8\%/+10.2\%$), LaMP-7 $+16.5\%/+16.9\%$ ($+12.8\%/+17.5\%$). Our absolute baselines differ from the paper's because of 4-bit quantization, the dev split and the history cap.

Reproduction fidelity tracks effect size. The task with the paper's largest effect (tweet paraphrasing) is where the reproduction is closest in magnitude, and tasks with small paper-reported effects diverge more. Small effects are more fragile to differences in model scale, quantization and data split. This pattern was derived by comparing tasks, not by adjusting the account of any single task.

B. Extension: the 2×2 ablation across runs

Table III gives absolute scores for the planned grid (single vs. multi-component, additive vs. NP) in every run. Absolute levels differ between runs (users, prompts, cleaning), so each run's gain is measured against its own no-steering baseline.

C. Run 5: refined pipeline

Run 5 is the cleanest test. Table IV and Fig. 2 show the results and Table V the settings selected on disjoint tuning users. The paper's method improves ROUGE-L on all three tasks ($+7.0\%$, $+21.8\%$, $+17.5\%$; paper $+3.2\%$, $+25.8\%$, $+12.8\%$) and METEOR on LaMP-5 ($+10.9\%$ against $+10.2\%$). On LaMP-4 the relative METEOR gain ($+150\%$) comes from a low unsteered score (0.057; steered 0.144). On LaMP-7 the METEOR gain is $+2.3\%$ against the paper's $+17.5\%$. The three additive conditions (Mean Diff, per-user PCA, shared basis) differ by at most 0.016 in ROUGE-L and 0.025 in METEOR on any task, while the two norm-preserving conditions behave differently: on LaMP-7 their ROUGE-L is at the unsteered level (0.299, 0.293 against 0.300).

Table VI gives paired comparisons against Mean Difference (Fig. 3). The shared-basis vector is inconclusive in all six comparisons, with differences of at most about 0.012 in absolute score; on LaMP-7 the difference is essentially zero (-0.0015, $[-0.0068, +0.0035]$). Both norm-preserving conditions are significantly lower than Mean Difference in three of six comparisons each (LaMP-7 in both metrics, LaMP-4 METEOR); the full method is never significantly better. Per-user PCA (k=3) equals Mean Difference in five of six comparisons and is better on LaMP-4 METEOR ($+0.0155$).

TABLE I
DESIGN OF THE FIVE EXTENSION RUNS (AS RECORDED IN THE SAVED SETTINGS AND OUTPUTS).

	Run 1	Run 2	Run 3	Run 4	Run 5 (final)
Test users / task	300	150 (149 LaMP-5)	300 (299 LaMP-5)	300 (298 LaMP-7)	300
Selection users	none (reproduction settings reused)	40 + 40 validation	60 validation	none (main run); 50 for sweeps	30 disjoint tuning users
Layers	16 (fixed)	12, 16, 20, 24	12, 16, 20, 24	16; sweep 12–24	16, 20
Strength	$\alpha = 0.25/0.5/0.5$	$\rho \in [0.05, 0.8]$	$\rho \in [0.05, 1.2]$	$\alpha = 0.02/0.02/0.5$; sweep $\times\{1, 4, 16, 64\}$	α per condition: $\{0.25..1.5\}$ plain, $\{0.5..4\}$ NP
Multi-component	per-user PCA, k=3	per-user PCA, $k \in \{3, 5\}$	per-user PCA	per-user PCA, $k \in \{1, 3, 5\}$	shared basis r=16 (+ per-user PCA control)
Output cleaning	none	yes	yes	none	yes
Controls	none	global, shrinkage	random direction	none	per-user PCA
Statistics	none	paired bootstrap	paired bootstrap	none	paired bootstrap vs. Mean Diff

TABLE II
REPRODUCTION VS. THE PAPER'S TABLE 2 ($n = 1{,}000$ PER TASK). Δ: RELATIVE GAIN OVER THE NON-PERSONALIZED BASELINE. PAPER Δ AS PRINTED.

Task	Metric	Paper			Ours		
		Base	Steer	Δ	Base	Steer	Δ
LaMP-4	ROUGE-L	.0398	.0411	+3.2%	.0988	.1020	+3.3%
LaMP-4	METEOR	.0790	.0809	+2.5%	.1172	.1207	+3.0%
LaMP-5	ROUGE-L	.1086	.1366	+25.8%	.1708	.2102	+23.1%
LaMP-5	METEOR	.2337	.2575	+10.2%	.2829	.3125	+10.5%
LaMP-7	ROUGE-L	.2506	.2827	+12.8%	.2525	.2942	+16.5%
LaMP-7	METEOR	.2588	.3042	+17.5%	.2770	.3239	+16.9%

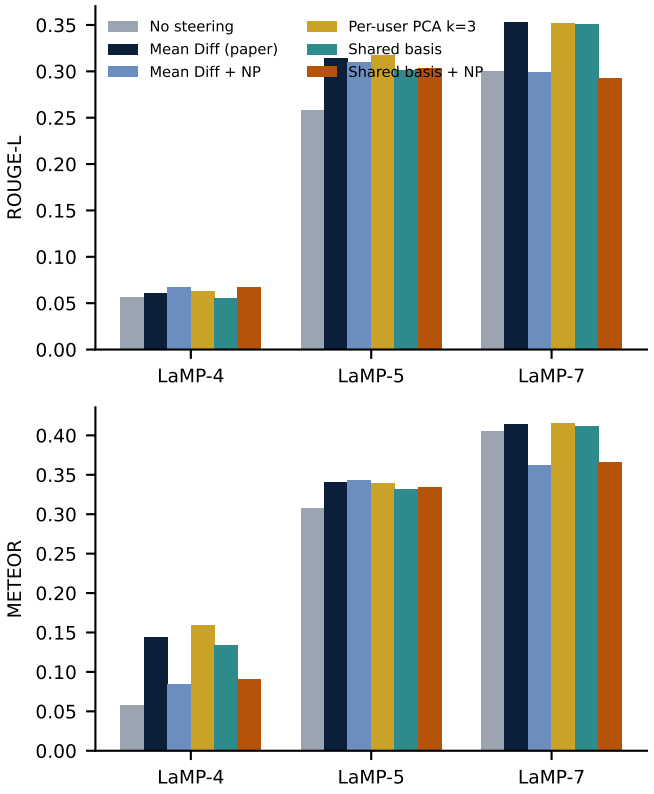


Fig. 2. Run 5: mean ROUGE-L (top) and METEOR (bottom) by condition.

D. Runs 1–4 and comparison with the paper

Fig. 4 and Table VII summarize the earlier runs. Run 1 (fixed settings, raw outputs) showed no measurable ROUGE-L change for the paper's method; the only significant gains were on LaMP-5 for the multi-component variants (+6.3% ROUGE-L and +3.8% METEOR for multi-PCA + NP). Run 2 (validation-tuned, cleaned) found a significant gain only on LaMP-7 METEOR (+4.0%). Run 3 found only LaMP-7 METEOR significant for the proposed method (+2.3%, lower bound near zero). Run 4 produced a large LaMP-7 effect (+29% ROUGE-L) that, after the same cleaning, falls to +12.5% (Table XIV).

VI. ABLATION STUDY

A. Ablations of the reproduced method

The reproduction went through 18 logged iterations. Key findings:

Steering strength. In an α sweep for one user with Qwen2.5-3B (layer 20), outputs were identical to the baseline for $\alpha \leq 0.5$, changed and coherent for $0.75 \leq \alpha \leq 1.5$, degraded at 2.0 and collapsed at 4.0 (Mean Diff scores 0.000).

Model and sample size. With Qwen2.5-3B, $n = 10$ per task gave mixed results (3 of 6 cells improved) and $n = 50$ gave +19.4% ROUGE-L on LaMP-4 but −38.9% on LaMP-7. We moved to LLaMA-2-7B-chat.

Grid search. A grid over layers $\{16, 20, 24, 28\}$ and $\alpha \in \{0.5, \ldots, 3.0\}$ on 10 validation users per task gave best cells 16/0.5, 24/1.0 and 16/0.5; the setting was then updated to 16/0.25, 16/0.5, 16/0.5 on a larger sample and kept for the final runs. A later robust search (three sets of 40 users, α from 0.25; selected values not recorded) and a LaMP-4 sensitivity study ($\alpha = 0.05$–0.25, within ±3%) did not change the picture.

Data split, prompt variants and history cap. Table VIII shows ROUGE-L gains by iteration. Gains varied strongly with the data split, with the prompt variant (bare, system or chat template, with generation length 15–40) and with the sample; for example, a per-task prompt choice gave −5.6% and −16.4% on LaMP-4/5. Raising the history cap to 60 changed little (−0.8%, +0.2%). Results stabilized in the clean final setup ($n = 300$ and $n = 1000$). A LongLaMP Topic Writing check on 38 users gave −6.5% ROUGE-L (paper +4.8%), and was not pursued.

TABLE III
THE 2 × 2 ABLATION GRID BY RUN. NS: NO STEERING; S/M: SINGLE/MULTI-COMPONENT; A/N: ADDITIVE/NORM-PRESERVING. FOR RUN 5, "MULTI" IS THE SHARED-BASIS VECTOR.

		ROUGE-L					METEOR				
Task	Run	NS	S-A	S-N	M-A	M-N	NS	S-A	S-N	M-A	M-N
LaMP-4	1	.0909	.0907	.0909	.0886	.0895	.1203	.1155	.1162	.1143	.1149
	2	.1105	.1156	.1136	.1136	.1127	.0933	.1006	.1001	.0981	.0983
	3	.1098	–	–	.1132	.1123	.0966	–	–	.1027	.1020
	4	.0971	.0963	.0963	.0961	.0959	.1206	.1207	.1202	.1200	.1198
	5	.0569	.0609	.0676	.0553	.0679	.0574	.1436	.0844	.1341	.0905
LaMP-5	1	.2137	.2203	.2222	.2261	.2271	.3179	.3131	.3160	.3284	.3300
	2	.3240	.3229	.3199	.3218	.3213	.3269	.3246	.3220	.3252	.3235
	3	.3196	–	–	.3205	.3197	.3348	–	–	.3346	.3340
	4	.2118	.2106	.2104	.2089	.2095	.3122	.3116	.3116	.3095	.3107
	5	.2578	.3141	.3100	.3019	.3036	.3076	.3411	.3425	.3321	.3345
LaMP-7	1	.3584	.3577	.3554	.3579	.3535	.3440	.3317	.3296	.3273	.3210
	2	.2975	.3016	.2978	.3042	.3035	.2640	.2746	.2716	.2754	.2759
	3	.2938	–	–	.2975	.2988	.2618	–	–	.2668	.2678
	4	.2429	.3133	.3041	.3132	.3081	.2644	.3100	.2977	.3086	.3051
	5	.3001	.3527	.2989	.3512	.2931	.4055	.4147	.3620	.4121	.3665

TABLE IV
RUN 5 RESULTS ($n = 300$ PER TASK): SCORE (RELATIVE CHANGE VS. NO STEERING, %).

	LaMP-4		LaMP-5		LaMP-7	
Condition	ROUGE-L	METEOR	ROUGE-L	METEOR	ROUGE-L	METEOR
No steering	.0569	.0574	.2578	.3076	.3001	.4055
Mean Diff (paper)	.0609 (+7.0)	.1436 (+150.2)	.3141 (+21.8)	.3411 (+10.9)	.3527 (+17.5)	.4147 (+2.3)
Mean Diff + NP	.0676 (+18.8)	.0844 (+47.0)	.3100 (+20.2)	.3425 (+11.3)	.2989 (-0.4)	.3620 (-10.7)
Per-user PCA, k=3	.0636 (+11.8)	.1590 (+177.0)	.3178 (+23.3)	.3392 (+10.3)	.3522 (+17.4)	.4148 (+2.3)
Shared basis	.0553 (-2.8)	.1341 (+133.6)	.3019 (+17.1)	.3321 (+8.0)	.3512 (+17.0)	.4121 (+1.6)
Shared basis + NP(full method)	.0679 (+19.3)	.0905 (+57.7)	.3036 (+17.8)	.3345 (+8.7)	.2931 (-2.3)	.3665 (-9.6)
Paper (StyleVector)	– (+3.2)	– (+2.5)	– (+25.8)	– (+10.2)	– (+12.8)	– (+17.5)

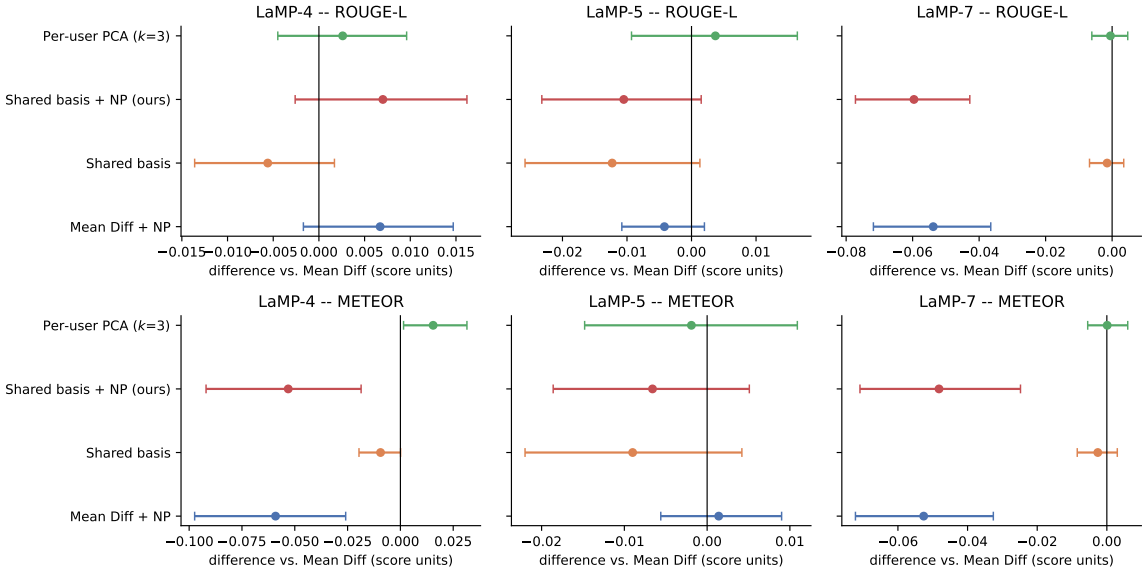


Fig. 3. Run 5 paired differences against Mean Difference with 95% intervals. Points right of zero are better than the paper's method.

Diagnostics. Baseline outputs were verbose: 97% (LaMP-4) and 100% (LaMP-5) began with a preamble, and averaged 22.4 and 24.2 words against 9.6 and 9.4 in the references. A ceiling effect was visible: the correlation between baseline score and gain was −0.36 (LaMP-4) and −0.28 (LaMP-5); users in the weaker half gained +0.009/+0.024 ROUGE-L while those in the stronger half changed by −0.019/−0.022. History items over 512 tokens were rare (0–0.3%).

B. Ablations of the extensions

Scorecards. Table IX counts significant paired comparisons. Across Runs 1–4, the paper's method beat no steering in 3 of 18 comparisons (worse in 1); norm preservation was

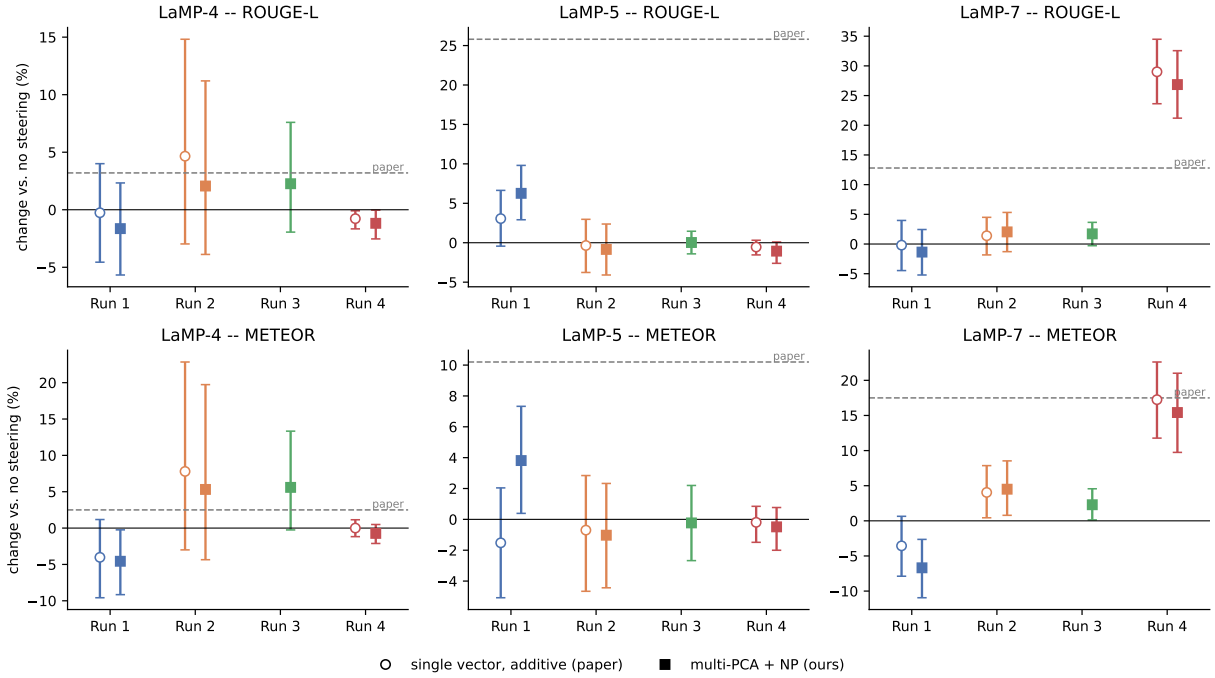


Fig. 4. Change vs. no steering for the paper's method (open circles) and the full method (filled squares), Runs 1–4, with 95% intervals. Dashed lines: the paper's reported gains. Run 4 is scored on raw outputs.

TABLE V
RUN 5 SELECTED LAYER / α PER CONDITION ON THE TUNING USERS.

Condition	LaMP-4	LaMP-5	LaMP-7
Mean Diff	16 / 1.5	20 / 1.0	20 / 0.25
Mean Diff + NP	16 / 1.0	20 / 1.0	16 / 0.5
Per-user PCA	16 / 1.5	16 / 0.25	20 / 0.25
Shared basis	16 / 1.5	16 / 1.0	20 / 0.25
Shared basis + NP	16 / 1.0	20 / 1.0	16 / 0.5

significantly better in 2 of 42 and worse in 3; multi-PCA was better in 1 of 18 and worse in 1; and the full method beat the paper's method in 1 of 18. Run 5 adds 24 comparisons against Mean Difference with the same pattern.

Number of components k (Run 4). Table X gives the sweep. Increasing k did not help: on LaMP-5, k=5 is significantly worse than k=1 in ROUGE-L (-0.9%) and k=3 is significantly worse in METEOR (-0.6%); no comparison between values of k is significantly positive. On LaMP-7 every variant improves by about $+29\%$ ROUGE-L regardless of k, and NP is significantly lower than additive (-3.8% for the single vector, -3.5% for k=5).

Layer sweep (Run 4, $n = 50$). On LaMP-7 the single additive vector improves over no steering at layers 16, 20 and 24 but not at 12, and multi-PCA + NP scores below the single additive vector at each layer (differences not significant). On LaMP-4 and LaMP-5 no layer changes either metric by more than 0.005.

Steering strength (Run 4, Fig. 5). On LaMP-7 a $4\times$ multiplier ($\alpha = 2$) already destroys the output (ROUGE-L about 0.03 for all four methods) and NP does not prevent this. At $64\times$ on LaMP-4/5 the multi-component vector degrades less than the single vector (ROUGE-L 0.087/0.094 for k=3 additive/NP against 0.078/0.083 on LaMP-4; 0.200/0.190 against 0.178/0.167 on LaMP-5), but the paired differences are not significant at $n = 50$. In validation (Runs 2–3), the selected ρ was the grid minimum (0.05) in all 24 selections; quality is roughly flat for $\rho \leq 0.2$ and collapses at $\rho = 1.2$, with NP and additive nearly identical.

Controls (Runs 2–3). Table XI shows that the global vector is statistically indistinguishable from the user vector on all tasks (numerically higher on LaMP-7: $+5.9\%$ METEOR over no steering, significant), shrinkage adds nothing beyond the global vector, and PCA + NP is not significantly better than a random direction (one borderline case, LaMP-4 METEOR, lower bound at zero). The diagnostic in Table XII shows why: on LaMP-7, 95% of user vectors have cosine above 0.9 with the global vector.

Output cleaning (the format confound). Table XIII shows that 93–100% of raw LaMP-4/5 generations start with a preamble in every run and that cleaning alone raises baseline ROUGE-L by 14–18% (LaMP-4), 51% (LaMP-5) and 19–21% (LaMP-7), far more than any steering effect in Runs 1–4. In Run 4 on LaMP-7, strong steering reduced chatty outputs from 85% to 21%; Table XIV rescored Run 4 with the same cleaning (ROUGE-L only): the LaMP-7 gain of the single vector falls from $+29\%$ to $+12.5\%$ (still significant), the multi-component NP variant is not better (-2.6% vs. single), and LaMP-4/5 are unchanged or slightly negative.

VII. DISCUSSION

Does the reproduction hold? Yes. At $n = 1{,}000$ the reproduction improves every task–metric pair with gains close

TABLE VI
RUN 5 PAIRED DIFFERENCES AGAINST MEAN DIFFERENCE (SCORE UNITS, 95% INTERVAL). BOLD: INTERVAL EXCLUDES ZERO.

Task	Configuration − Mean Diff	ROUGE-L: diff [95% CI]	METEOR: diff [95% CI]
LaMP-4	Mean Diff + NP	+0.0067 [-0.0017,+0.0147]	**-0.0591 [-0.0974,-0.0259]**
	Shared basis	-0.0056 [-0.0136,+0.0017]	-0.0094 [-0.0196,+0.0001]
	Shared basis + NP	+0.0070 [-0.0026,+0.0162]	**-0.0531 [-0.0920,-0.0186]**
	Per-user PCA, k=3	+0.0026 [-0.0045,+0.0096]	**+0.0155 [+0.0015,+0.0315]**
LaMP-5	Mean Diff + NP	-0.0042 [-0.0108,+0.0020]	+0.0014 [-0.0056,+0.0090]
	Shared basis	-0.0123 [-0.0258,+0.0013]	-0.0090 [-0.0220,+0.0042]
	Shared basis + NP	-0.0105 [-0.0232,+0.0015]	-0.0066 [-0.0186,+0.0051]
	Per-user PCA, k=3	+0.0037 [-0.0093,+0.0164]	-0.0019 [-0.0148,+0.0109]
LaMP-7	Mean Diff + NP	**-0.0538 [-0.0718,-0.0365]**	**-0.0526 [-0.0722,-0.0326]**
	Shared basis	-0.0015 [-0.0068,+0.0035]	-0.0026 [-0.0085,+0.0030]
	Shared basis + NP	**-0.0596 [-0.0772,-0.0428]**	**-0.0482 [-0.0709,-0.0248]**
	Per-user PCA, k=3	-0.0005 [-0.0061,+0.0047]	+0.0001 [-0.0055,+0.0060]

TABLE VII
RELATIVE GAIN (%) OF THE PAPER'S METHOD OVER ITS OWN BASELINE, ROUGE-L / METEOR. †: 95% INTERVAL EXCLUDES ZERO (RUNS 1–4). RUN 3 IS PER-USER PCA (ADDITIVE); RUN 4 IS RAW-SCORED. RUN 5 HAS NO INTERVALS AGAINST NO STEERING.

Source	LaMP-4	LaMP-5	LaMP-7
Paper	+3.2 / +2.5	+25.8 / +10.2	+12.8 / +17.5
Reproduction (n=1000)	+3.3 / +3.0	+23.1 / +10.5	+16.5 / +16.9
Run 1	-0.3 / -4.0	+3.1 / -1.5	-0.2 / -3.6
Run 2	+4.7 / +7.8	-0.3 / -0.7	+1.4 / +4.0†
Run 3	+3.1 / +6.3	+0.3 / -0.1	+1.3 / +1.9
Run 4	-0.8† / +0.0	-0.5 / -0.2	+29.0† / +17.2†
Run 5	+7.0 / +150.2	+21.8 / +10.9	+17.5 / +2.3

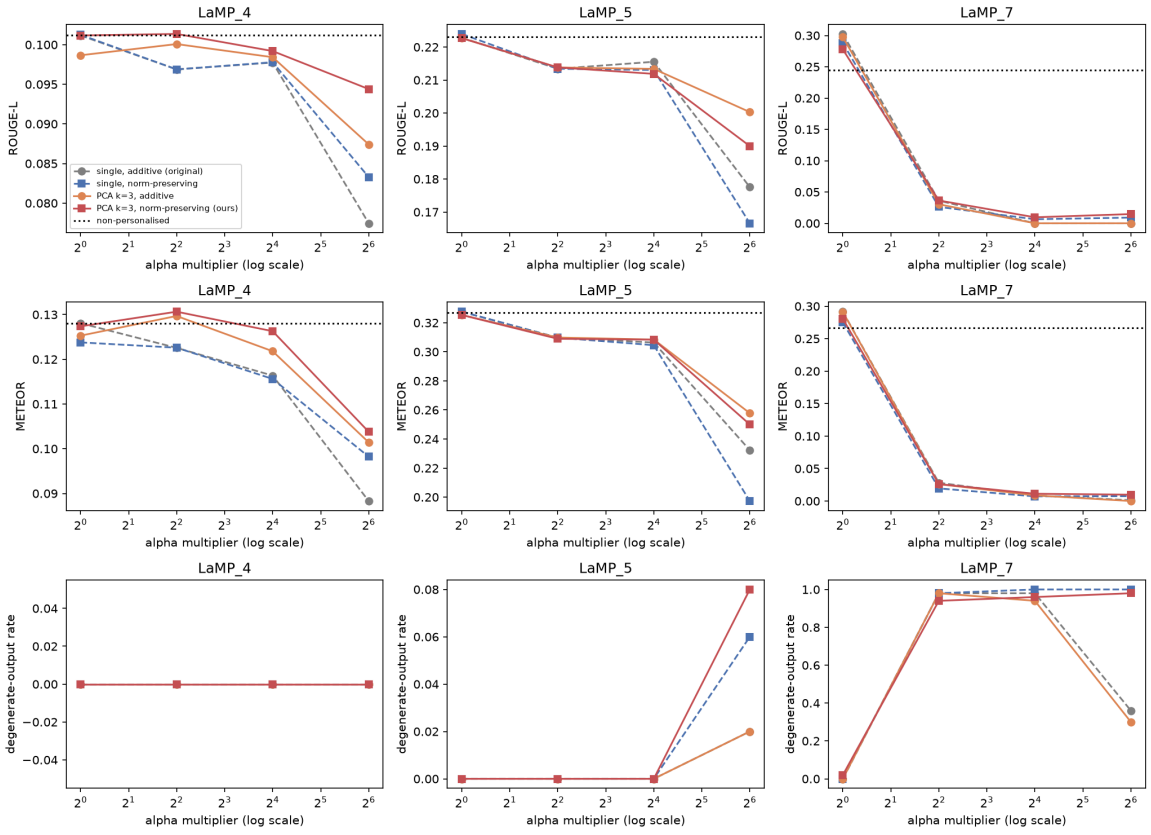


Fig. 5. Run 4 strength sweep (original figure): ROUGE-L, METEOR and degenerate-output rate vs. strength multiplier.

to the paper's, and Run 5 reproduces the large LaMP-5 gain (+21.8%/+10.9%) and improves ROUGE-L on LaMP-4 and LaMP-7. Earlier runs did not, and the ablations explain why: the result depends on the data split, prompt variant, sample size and especially on the output format. The Run 5 pipeline changed several things at once (cleaning, history construction,

TABLE VIII
REPRODUCTION ABLATION: ROUGE-L GAIN (%) BY ITERATION (LLAMA-2-7B-CHAT).

Configuration	LaMP-4	LaMP-5	LaMP-7
Paper	+3.3	+25.8	+12.8
Grid-searched layer/α, $n \approx 50$	+3.1	-4.4	+24.8
Shuffled dev split, $n \approx 201$	+1.6	+2.6	+29.2
One prompt variant for all, $n \approx 200$	-0.4	+6.2	-9.4
Prompt variant per task, $n \approx 200$	-5.6	-16.4	+29.2
Larger sample, $n \approx 800$	-2.2	-2.2	+26.4
Robust grid search, $n \approx 800$	-4.5	+0.6	+26.4
All dev users, $n \approx 806$	-0.8	+0.6	+26.4
History cap 60, $n \approx 517/149$	-0.8	+0.2	–
Clean setup, $n = 300$	+4.8	+29.8	+16.4
Clean setup, $n = 1000$	+3.3	+23.1	+16.5

TABLE IX
SCORECARD OF SIGNIFICANT PAIRED COMPARISONS (ROUGE-L AND METEOR, THREE TASKS).

Question	N	Better	Worse	Incl.
Runs 1–4				
Paper's method vs. no steering	18	3	1	14
NP on single vector	18	2	2	14
NP on multi-PCA	24	0	1	23
Multi-PCA vs. single (additive)	18	1	1	16
Full method vs. paper's method	18	1	0	17
Run 5 (vs. Mean Diff)				
NP on single vector	6	0	3	3
Shared basis vs. single	6	0	0	6
Full method vs. paper's method	6	0	3	3
Per-user PCA (k=3) vs. single	6	1	0	5

matched prompts, all-position steering, per-condition tuning), so it does not isolate which change accounts for the improvement.

Do the extensions help? In these experiments, no. The shared-basis vector equals Mean Difference in 6 of 6 comparisons; per-user PCA equals it in 5 of 6; norm preservation scores lower in 3 of 6 comparisons for each NP condition, even with strengths up to 4 on separate tuning users, and none selected more than 1.0. The proposal's success criterion (the full method better than the single-vector baseline on at least two of three tasks) is therefore not met. The tuning-user scores show the same ordering as the evaluation users (for example LaMP-4: 0.085 and 0.083 for NP against 0.134 and 0.136 additive).

Why not? Several observations are consistent with the plain mean already capturing most of the usable signal. (i) On LaMP-7, user vectors are nearly identical to the global vector (Table XII), so projecting onto a shared basis changes little. (ii) Selected strengths were at the grid minimum in Runs 2–3, so NP was evaluated where it has little room to act (to first order it removes only the component of p parallel to h, at most $0.05\|h\|$ at $\rho = 0.05$). (iii) Where strength is large, quality has already collapsed and NP rescues it only partially. A possible explanation for the LaMP-7 NP deficit, not tested here, is that the effect relies on changing the activation magnitude, which NP removes by construction. On LaMP-4 the pattern of slightly higher ROUGE-L but much lower METEOR is what shorter outputs would produce, since METEOR is recall-weighted; output lengths would confirm this.

Is the gain personal? Partly unresolved. The global vector and a random direction perform about as well as the personal variants in Runs 2–3, and a shared-basis projection that discards user-specific variation loses nothing in Run 5. This indicates that much of what steering contributes on these benchmarks is shared across users (for example a "tweet register" or the removal of chat-style preambles) rather than user-specific.

Evaluation matters. Output cleaning alone changed baseline ROUGE-L by 14–51%, far more than any steering effect, and raw-output scoring can both create effects (LaMP-7, Run 4) and hide them. Identical cleaning across conditions and held-out tuning users are therefore essential for fair comparison.

VIII. LIMITATIONS

- **Absolute scores differ from the paper's**: 4-bit instead of 8-bit quantization (GPU memory), the public dev split instead of the paper's evaluation, a history cap, and a smaller GPU. Only direction and relative size are compared.
- **Single-reference metrics.** ROUGE-L and METEOR penalize a stylistically faithful paraphrase that uses different words, especially on tweets.
- **Statistical power.** Short targets and 150–300 users give wide intervals; effects of 1–3% are hard to detect, and no multiple-comparison correction is used.
- **Cleaning is a heuristic** that differs between runs; runs are comparable through their own baselines, not in absolute terms.
- **Run 5 summary data.** Run 5 is reported from the notebook's summary tables, so intervals exist for comparisons between steered conditions but not against no steering.
- **Combined pipeline changes** in Run 5 prevent attribution to a single fix.
- **Duplicate profiles.** Only 102 of the 150 LaMP-4 test users in Run 2 have distinct vector statistics, which suggests repeated profiles and reduces effective sample size.
- **Unrecorded settings.** The model identifier, and for some runs the history cap and PCA k, are not stored in the archives.
- **Scope.** Three LaMP tasks with one model; no long-form tasks (a small LongLaMP check was inconclusive), and no multi-domain users.

IX. CONCLUSION AND FUTURE WORK

We reimplemented StyleVector from the paper alone and reproduced its improvement on all six LaMP task–metric pairs at $n = 1{,}000$ users, using consumer hardware, and built a live demonstration system. We then evaluated multi-component style vectors and norm-preserving steering in a 2×2 ablation over five runs, with sweeps over k, layer and strength and with global, shrinkage and random controls. Neither extension improved on the paper's mean-difference steering; the shared-basis vector matched it and norm-preserving variants scored lower in half of the comparisons. The strongest configuration tested is Mean Difference steering with a tuned layer and

TABLE X
RUN 4 MAIN RESULTS (RAW OUTPUTS, $n = 300$): EFFECT OF k AND NP.

	LaMP-4		LaMP-5		LaMP-7	
Configuration	ROUGE-L	METEOR	ROUGE-L	METEOR	ROUGE-L	METEOR
No steering	.0971	.1206	.2118	.3122	.2429	.2644
Single (mean diff), additive	.0963	.1207	.2106	.3116	.3133	.3100
Single + NP	.0963	.1202	.2104	.3116	.3041	.2977
PCA k=1, additive	.0964	.1201	.2106	.3115	.3121	.3085
PCA k=1 + NP	.0965	.1206	.2104	.3113	.3039	.2988
PCA k=3, additive	.0961	.1200	.2089	.3095	.3132	.3086
PCA k=3 + NP	.0959	.1198	.2095	.3107	.3081	.3051
PCA k=5, additive	.0961	.1197	.2086	.3094	.3166	.3146
PCA k=5 + NP	.0965	.1202	.2086	.3093	.3082	.3084

TABLE XI
CONTROLS: PAIRED DIFFERENCES AS % OF THE NO-STEERING SCORE (ROUGE-L / METEOR).

Run	Comparison	LaMP-4	LaMP-5	LaMP-7
2	Global vs. user	-5.3 / -9.5	-0.4 / -0.6	+1.0 / +1.9
2	Shrunk vs. user	-3.0 / -6.0	-0.3 / -0.8	+0.6 / +1.0
2	Shrunk vs. global	+2.3 / +3.5	+0.0 / -0.2	-0.5 / -0.9
3	PCA+NP vs. random	+2.9 / +5.8	-0.4 / -0.1	+1.5 / +1.1
3	Random vs. none	-0.6 / -0.2	+0.4 / -0.1	+0.2 / +1.2

TABLE XII
RUN 2 DIAGNOSTIC: SIMILARITY BETWEEN EACH USER'S MEAN-DIFFERENCE VECTOR AND THE GLOBAL VECTOR.

Task	Users	Mean cosine	Users with cos $>$ 0.9 (%)
LaMP-4	150	0.562	56.7
LaMP-5	149	0.652	0.0
LaMP-7	150	0.950	95.3

TABLE XIII
RAW OUTPUTS STARTING WITH A PREAMBLE (%, NO STEERING) AND RELATIVE ROUGE-L GAIN FROM CLEANING THE UNSTEERED BASELINE.

	Preamble (%), Run				Cleaning gain (%)	
Task	1	2	3	4	Run 2	Run 3
LaMP-4	93.0	97.3	96.3	97.7	+17.5	+14.5
LaMP-5	99.7	100.0	100.0	100.0	+50.6	+51.3
LaMP-7	17.0	86.0	83.0	84.9	+20.6	+18.9

TABLE XIV
RUN 4 RESCORED WITH OUTPUT CLEANING (ROUGE-L; RELATIVE CHANGE [95% CI]).

Task	None	Single	Single vs. none	Multi+NP vs. none
LaMP-4	.1144	.1140	-0.3 [-1.1,+0.4]	-1.0 [-2.7,+0.2]
LaMP-5	.3209	.3195	-0.4 [-1.5,+0.5]	-1.2 [-2.6,-0.0]
LaMP-7	.2910	.3273	+12.5 [+7.8,+17.1]	+9.9 [+5.0,+14.6]

strength. The ablations also show that output formatting, data split and prompt choice can matter as much as the method.

Future work. (i) Per-user results with intervals against no steering for the refined pipeline, and a controlled study that changes one pipeline element at a time. (ii) A global-vector control in the refined pipeline, to quantify what is personal. (iii) Testing whether NP changes output length and whether the LaMP-7 effect depends on activation magnitude. (iv) Separating shared from user-specific style, for example by learning the shared subspace and then steering only with the residual, or by sparse feature decompositions [16]. (v) Task-specific selection of recipes (algorithm, layer, strength) on held-out users, with a single evaluation on untouched test users. (vi) Larger models, long-form tasks and cross-domain histories.

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APPENDIX A
AI PROMPTS GIVEN TO THE LLM

An AI assistant (Anthropic's Claude, used through claude.ai) was used throughout. In the reproduction phase it proposed the implementation of the paper's method, wrote the notebook and web-application code, helped diagnose errors and drafted the report. In the extension phase it reviewed the earlier notebook, proposed the corrected pipeline used in Run 5, wrote the analysis scripts, report, presentation and README, and drafted this document. The author ran all experiments on the lab GPU machine, relayed errors and outputs, made the scoping decisions (task selection, abandoning LongLaMP as infeasible, accepting 4-bit quantization), noticed the mismatch between a live generation and its aggregate score that exposed a bug, supplied the web-page redesign, verified source files when numbers disagreed, and reviewed all results. Other platforms: Google Colab (considered for initial setup), Jupyter notebooks on the lab machine, Flask for the demonstration system. The prompts below are the author's requests, preserved as made (dictated requests lightly cleaned of filler words).

A.1 Reproduction phase

1) "Can you access the LaMP dataset used in the paper? If so, explain how to download it and give me code to load it."
2) "I want to run this on Google Colab. What are the steps?"
3) "Loading the model with load_in_8bit=True raises a TypeError. What should I change?"
4) "The model now loads correctly. Implement the paper's method so its results can be compared against the paper's Table 2." (Table 2 attached as an image.)
5) "The evaluation cell runs very slowly and I had to interrupt it. Why, and how do I fix it?"
6) "The evaluation cell now fails with this error. What should I change?" (repeated for chat-template and NLTK-resource errors.)
7) "Are these outputs what I should expect, or does the logic need to change?" (after steered outputs degenerated at high α.)
8) "Give me a ready-made notebook with numbered, described cells that loads the dataset, runs the paper's Mean Difference method on the full data with tables and plots like the paper's Table 2."
9) "My results don't match the paper's. What should I do?"
10) "I want the results table to look like the paper's so comparison is easy. Can we also use LLaMA-2-7B instead of Qwen2.5-3B?"
11) "I need more accurate numbers and have time. What should I do to improve accuracy? Also, how can I view the imported dataset as a table?"
12) "Can we use other datasets from the paper instead of LaMP-4 and LaMP-5?" (the LongLaMP attempt, later abandoned.)
13) "The notebook has become cluttered from testing, including an unnecessary grid search. Give me a clean version."
14) "Using this model, build a website that lists users from the dataset, lets me enter a prompt, and shows the base and StyleVector outputs with analytics."
15) "Star the users with the largest improvement so I can pick them while presenting."
16) "The benchmark reports roughly 16–28% improvement on LaMP-7, but a live generation shows about 0.5%. How is that possible?"
17) "Extend the web app into a presentation deck with animations, keyboard navigation, and the live demo embedded as one slide."
18) "Modify the UI of both pages and present the improvement data properly."
19) "When I test my own prompt, the improvement statistics are missing. Add them."
20) "The Generate button shows no output and keeps loading. Fix it."
21) "Write the midterm report in LaTeX."

A.2 Extension phase

1) Go through the project proposal and the end-term notebook; check whether the code is written correctly; if not, correct it and try to get a better result than the paper's; either explain why I am not getting improved results or give the correct code.
2) I have already run my code four times. Go through all four checkpoint zips (end-term, v2, v3, extension) and prepare an analytics report in LaTeX that includes all the data from these ablations.
3) (With the Run 5 output pasted.) "I got this output for the final run. Now update the LaTeX PDF and explain everything properly."
4) (With the best-configuration file.) Take the data from this and update the PDF; before the final report, ask me for the incomplete parts; include my name and ID. *(The assistant kept all reported findings and asked for the missing data.)*
5) Give me the LaTeX code as well so that I can edit it.
6) Create a notebook that combines the settings that worked in earlier runs (layer, α, algorithm) per task, without re-running the paper's method. *(A selection notebook with held-out tuning users was produced; it has not been run, and no results from it are reported.)*
7) Prepare a final presentation (more than three slides) from the earlier report, the presentation and the run results: problem statement, architecture, ablations of the paper's method, the final result, then the ablations of the modifications (Runs 1–5); do not say "midterm" and do not include Run 6.
8) (Three follow-ups on the deck.) Make the design more formal, visually appealing and catchy; fix the empty space and constrained text; format the equations on the architecture slide correctly.
9) Write a README.md for the end-term submission with the given folder structure.
10) According to the given LaTeX template and the two reports of phase 2 and phase 3, prepare a final report with the following sections: title, abstract, introduction, related work, proposed methodology, experimental setup, results and analysis, ablation study, discussion, limitations, conclusion and future work, references, and an appendix with the AI prompts given to the LLM.